Cafe Wi-Fi Network

Using Cisco Packet Tracer

Project Report Submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

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1. ABSTRACT

This project demonstrates the design and simulation of a small café network using Cisco Packet Tracer. The main objective is to provide both wired and wireless internet connectivity for customers and café staff in a simple and efficient manner. The network includes a wireless router, an access point, a switch, customer devices such as laptops and smartphones, and a cashier computer for business operations. Dynamic Host Configuration Protocol (DHCP) is used to automatically assign IP addresses to all connected devices, ensuring easy network management and reducing configuration errors. The wireless network allows customers to access the internet conveniently while the wired connection provides stable connectivity for the café's internal system. This simulation represents a real-world café networking environment and demonstrates basic networking concepts such as IP addressing, wireless communication, and network connectivity. The project highlights how small businesses can implement cost-effective and reliable network solutions using simple network infrastructure. The proposed network design reflects a real-world café environment and demonstrates fundamental networking concepts including wireless communication, IP addressing, and basic network configuration. This project highlights a simple, cost-effective, and practical networking solution suitable for small businesses

2. DESIGN

2.1 Project Objective

The main objective of this project is to design and simulate a Café Wi-Fi Network using Cisco Packet Tracer.

The goals of the project are:

- To provide wireless internet access for customers

- To provide wired connectivity for café staff (cashier system)
- To implement DHCP (Dynamic IP addressing)
- To ensure secure Wi-Fi communication using WPA2-PSK
- To create a simple, cost-effective network design for small businesse

2.2 Network Topology Used

Devices Used

The following devices are used in the network:

- Router (1941)
- Switch (2960)
- Access Point
- Desktop PC (Cashier System)
- Laptops (Customer devices)
- Smartphone
- Ethernet cables

Topology Structure

The project uses a **Star Topology**, where:

- The **router** acts as the central device
- The **switch** connects wired devices
- The **access point** provides wireless connectivity
- All devices communicate through the central network

2.3 IP Addressing Scheme

Device	IP Address
Router	192.168.1.1
PC (Cashier)	DHCP Assigned
Laptop	DHCP Assigned
Smartphone	DHCP Assigned

- Subnet Mask: **255.255.255.0**
- Default Gateway: **192.168.1.1**

2.4 Design Explanation

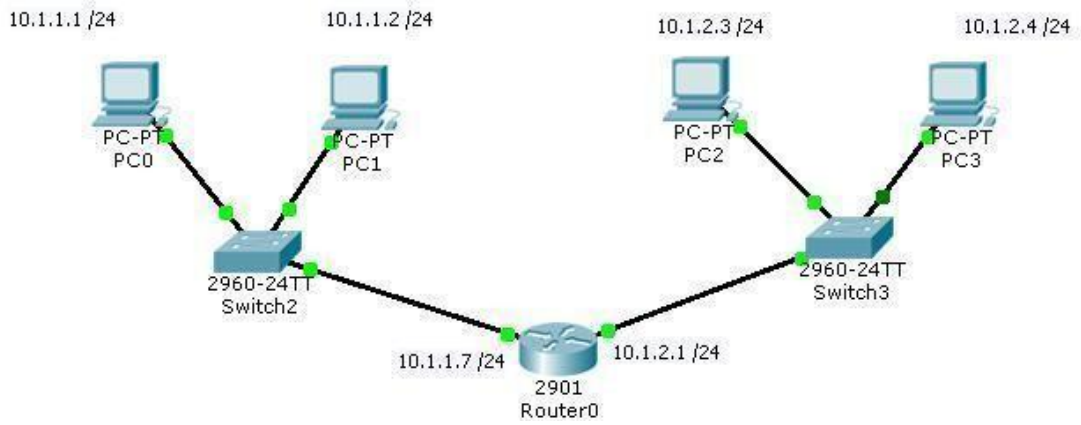
The network is designed to simulate a real café environment.

- The **router** is configured with DHCP to automatically assign IP addresses
- The **switch** connects the cashier PC using a wired connection for stable performance
- The **access point** provides wireless connectivity to customer devices
- The **SSID (network name)** is set as *cafe_wifi*
- Security is implemented using **WPA2-PSK password authentication**

This design ensures:

- Easy network management
- Secure wireless communication
- Reliable connectivity for both staff and customers

3. DEVELOPMENT



3.1 Device Placement

Devices are placed in the workspace as follows:

- Router placed at the center/top
- Switch placed near the router
- Access Point placed near the switch
- PC connected to switch
- Laptops and smartphone placed around access point

This arrangement makes the topology **clear and easy to understand**

3.2 Physical Connections

- The following connections are established:
- Router (G0/0) → Switch (Fa0/1)

- Switch (Fa0/2) → PC (FastEthernet0)
- Switch (Fa0/3) → Access Point (Port 0)
- Connection type used:
- **Copper Straight-Through Cable**
- Wireless connections:
- Laptop → Access Point
- Smartphone → Access Point

3.3 Verification

The network is verified using the following steps:

1. **All devices are set to DHCP mode**
2. **Devices receive IP addresses automatically**
3. **Connectivity is tested using the ping command**
4. Example:
5. `ping 192.168.1.1`
6. Successful output:
7. `Reply from 192.168.1.1`

4. <u>COMMAND CODES</u>

4.1 PC CONFIGURATION

PC0

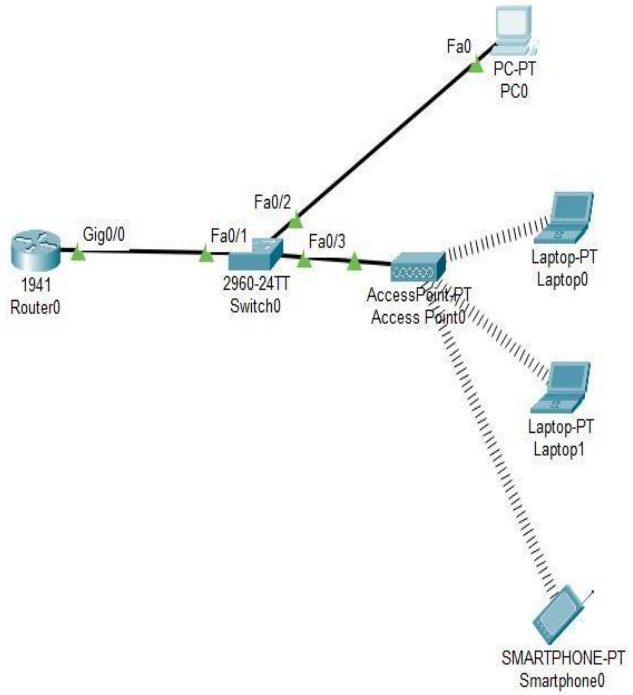
- IP Address: 192.168.1.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

PC1

- IP Address: 192.168.2.2

- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1

5. EXECUTION SCREENSHOT



PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Laptop0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time=32ms TTL=128
Reply from 192.168.1.4: bytes=32 time=27ms TTL=128
Reply from 192.168.1.4: bytes=32 time=27ms TTL=128
Reply from 192.168.1.4: bytes=32 time=17ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 17ms, Maximum = 32ms, Average = 25ms

C:\>ping 192.168.1.5

Pinging 192.168.1.5 with 32 bytes of data:

Reply from 192.168.1.5: bytes=32 time=38ms TTL=128
Reply from 192.168.1.5: bytes=32 time=23ms TTL=128
Reply from 192.168.1.5: bytes=32 time=32ms TTL=128
Reply from 192.168.1.5: bytes=32 time=24ms TTL=128

Ping statistics for 192.168.1.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 23ms, Maximum = 38ms, Average = 29ms

C:\>
```

Laptop1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=21ms TTL=128
Reply from 192.168.1.3: bytes=32 time=28ms TTL=128
Reply from 192.168.1.3: bytes=32 time=23ms TTL=128
Reply from 192.168.1.3: bytes=32 time=27ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 21ms, Maximum = 28ms, Average = 24ms

C:\>ping 192.168.1.5

Pinging 192.168.1.5 with 32 bytes of data:

Reply from 192.168.1.5: bytes=32 time=54ms TTL=128
Reply from 192.168.1.5: bytes=32 time=25ms TTL=128
Reply from 192.168.1.5: bytes=32 time=31ms TTL=128
Reply from 192.168.1.5: bytes=32 time=34ms TTL=128

Ping statistics for 192.168.1.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 25ms, Maximum = 54ms, Average = 36ms

C:\>
```